Hints: Number theory – 3

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Euler: with a primitive root g, $a = g^k$ is a square if and only if k is even, if and only if $a^{(p-1)/2} = g^{k(p-1)/2} = 1$. Otherwise $a^{(p-1)/2} = -1$, the only other square root of 1. Multiplicativity follows at once, and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

Hint 2. Infinitely many $p \equiv 1 \pmod 4$: every odd prime factor p of $(2N!)^2 + 1$ has $\left(\frac{-1}{p}\right) = 1$ and $p > N$.

Hint 3. Gauss's lemma: write each ka, $1 \leqslant k \leqslant \frac{p-1}{2}$, as $\pm r_k$ with $1 \leqslant r_k \leqslant \frac{p-1}{2}$. The r_k are a permutation of $1, \dots, \frac{p-1}{2}$ (if $r_k = r_l$ then $k \equiv \pm l$). Multiply everything: $a^{(p-1)/2} \equiv (-1)^{\mu}$. For $a = 2$, count the k with $p/4 < k < p/2$ and check the four classes of p mod 8.

Problem 2

Hint 1. By Gauss's lemma (in Eisenstein's form) $\left(\frac{q}{p}\right) = (-1)^{\sum_{k=1}^{(p-1)/2} \lfloor kq/p \rfloor}$ for odd q. This exponent counts the lattice points under the diagonal of the rectangle $(0, p/2) \times (0, q/2)$. The other symbol counts those above, and no lattice point lies on the diagonal.

Hint 2. $219 = 3 \cdot 73$. Use $\left(\frac{3}{383}\right) = -\left(\frac{383}{3}\right) = -\left(\frac{2}{3}\right) = 1$ and $\left(\frac{73}{383}\right) = \left(\frac{383}{73}\right) = \left(\frac{18}{73}\right) = \left(\frac{2}{73}\right) = 1$. So the congruence is solvable.

Hint 3. $\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}\left(\frac{p}{3}\right)$. Answer: $p \equiv \pm 1 \pmod{12}$.

Problem 3

Hint 1. The task is to find n with $f(n) = n^5 + n^4 - 3$ a non-residue mod p. Small n settle many primes: $f(1) = -1$ handles $p \equiv 3 \pmod 4$, and $f(0) = -3$ handles $p \equiv 2 \pmod 3$. No finite list of fixed n works for all p, so a counting argument is needed.

Hint 2. If $f(n)$ is a square or 0 for every n, then $\sum_n \left(\frac{f(n)}{p}\right) \geqslant p - 5$. So the curve $y^2 = f(x)$ has at least $2p - 5$ affine points.

Hint 3. f has odd degree 5, so it is not a constant times a square. Weil's bound for this genus-2 curve gives $\left|\sum_n \left(\frac{f(n)}{p}\right)\right| \leqslant 4\sqrt{p}$. So $p - 5 \leqslant 4\sqrt{p}$, i.e. $p \leqslant 25$. The remaining primes are covered by Hint 1 except $p = 13$, where $f(2) = 45 \equiv 6$ is a non-residue.

Problem 4

Hint 1. Multiplicity of the root 1: apply the operator $\theta = x\frac{d}{dx}$ repeatedly and evaluate at $x = 1$. Because the multiplicity is $< p$, $(x-1)^e \mid q$ if and only if $(\theta^j q)(1) = 0$ for all $j < e$.

Hint 2. $(\theta^j q)(1) = \sum_{k=1}^{p-1} k^{\frac{p-1}{2}+j}$, which is 0 in $\mathbb{F}_p$ unless $(p-1) \mid \frac{p-1}{2} + j$. For $j = \frac{p-1}{2}$ it equals -1. Answer: $(x-1)^{(p-1)/2}$.

Problem 5

Hint 1. Thue: for $x^2 \equiv -1 \pmod p$, consider the numbers $u - xv$ with $0 \leqslant u, v < \sqrt{p}$. By pigeonhole two of them coincide mod p, which gives $a \equiv xb$ with $0 < a^2 + b^2 < 2p$ and $p \mid a^2 + b^2$.

Necessity: squares are $0, 1 \pmod 4$, and a prime $q \equiv 3 \pmod 4$ dividing $a^2 + b^2$ divides both a and b, so it occurs to an even power.

Hint 2. In $\mathbb{Z}[i]$ (Euclidean for the norm): $p \mid (x + i)(x - i)$ but p divides neither factor, so p is not prime in $\mathbb{Z}[i]$, and $p = \pi\bar{\pi} = a^2 + b^2$.

Hint 3. Counting: $r_2(n)$ is the number of $\alpha \in \mathbb{Z}[i]$ with $N(\alpha) = n$. Use unique factorisation and count the choices for each prime type. Answer: $r_2(n) = 4\big(d_1(n) - d_3(n)\big)$, where d_j counts the divisors $\equiv j \pmod 4$.